

1-Which of the following is not an enzyme involved in glycolysis?

- A. Aldolase
- B. Glucose oxidase**
- C. Enolase
- D. Hexokinase

2-Creatinine is the product of..... creatine or phosphocreatine in muscles.

- A. Oxidation-Reduction
- B. Dehydration-Dephosphorylation**
- C. Reduction-Phosphorylation
- D. Hydration-Dehydration

3-Cori's cycle transfers?

- A. Lactate from liver to muscles
- B. Pyruvate from liver to muscles
- C. Glucose from muscles to liver
- D. Lactate from muscles to liver**

4-Which amino acids are found in urine in cystinuria?

- A. Threonine-Asparagine-Glutamine-Histidine
- B. Lysine-Arginine-Ornithine-Cystine**
- C. Tyrosine-Phenylalanine-Glycine-Leucine
- D. Glutamate-Aspartate-Serine-Alanine

5-High level of uric acid in blood is also called?

- A. Azotemia
- B. Uricemia**
- C. Uremia
- D. BUN

6-Glycogen is converted to glucose-1-phosphate by?

- A. Phosphatase
- B. Branching enzyme
- C. UDPG transferase
- D. Phosphorylase**

7-Two important byproducts of the Hexose monophosphate (HMP) shunt are?

- A. Pentose sugars and 4 membered sugars
- B. Pentose sugars and sedoheptulose
- C. NADH and pentose sugars
- D. NADPH and pentose sugars**

8-Which amino acid is not required for creatine synthesis?

- A. Glycine
- B. Ornithine**
- C. Methionine
- D. Arginine

9-Conversion of glucose to glucose-6-phosphate in human liver is catalyzed by.....?

- A. Hexokinase and glucokinase
- B. Glucose-6-phosphate dehydrogenase
- C. Hexokinase only
- D. Glucokinase only

10-Carbamoyl phosphate synthetase I regulates the urea cycle. Which of the following compounds activates this enzyme?

- A. Fumarate
- B. Argininosuccinate
- C. Carbamoyl phosphate
- D. N-acetyl glutamate

11-The reaction succinyl COA to succinate requires?

- A. GDP
- B. ADP
- C. NADP+
- D. CDP

12-Which of the following enzyme is required for the hexose monophosphate (HMP) shunt pathway?

- A. Glucose-6-phosphate dehydrogenase
- B. Aldolase
- C. Glucose-6-phosphatase
- D. Phosphorylase

13-From which amino acid is carnitine obtained?

- A. Glutamate
- B. Lysine**
- C. Ornithine
- D. Tryptophan

14-Which vitamin is needed to convert serine to glycine?

- A. C
- B. B2
- C. B9**
- D. D

15-Which enzyme is missing in Alkaptonuria?

- A. Fumarylacetoacetate hydrolase (FAH)
- B. Tyrosine aminotransferase (TAT)
- C. Homogentisate oxidase (HGD)**
- D. Phenylalanine hydroxylase (PAH)

16-Which of the following is an enzyme required for glycolysis?

- A. Pyruvate kinase**
- B. Pyruvate carboxylase
- C. Glycerokinase
- D. Malate dehydrogenase

17-Which amino acid is needed in the urea cycle to convert Citrulline to Argininosuccinate?

A. Aspartate

B. Glutamine

C. Lysine

D. Glycine

18-Which enzyme plays an important role in the final stage of purine base catabolism to uric acid?

A. Xanthine oxidase

B. Guanine deaminase

C. Nucleotidase

D. Adenosine deaminase

19-Which enzyme converts chymotrypsinogen to chymotrypsin?

A. Carboxy peptidase

B. Elastase

C. Trypsin

D. Enteropeptidase

20-Which one of the following is the major blood NPN?

A. Creatinine

B. Urea

C. Uric acid

D. Ammonia

21-Which of the following is correct about decreasing fat absorption?

- A. Ezetimibe decreases absorption of dietary cholesterol
- B. Orlistat inhibits gastric lipases
- C. Humans metabolize the cholesterol ring to carbon dioxide and water
- D. Orlistat inhibits pancreatic lipases

22-The acetyl CoA is produced in the mitochondria and must be transported into cytosol for synthesis of fatty acid. Which of the following is true regarding its transport?

- A. Acetyl CoA is transported by its specific transporter protein
- B. Acetyl CoA is converted into pyruvate, enters into cytosol and acetyl CoA is regenerated
- C. Acetyl CoA is converted into citrate, enters into cytosol and acetyl CoA is regenerated.
- D. Acetyl CoA is diffused from mitochondrial membrane

23-Net generation of energy on complete oxidation of a 15-carbon fatty acid is.....?

- A. Equal to the energy generation from a 16-carbon fatty acid
- B. Equal to the energy generation from a 14-carbon fatty acid
- C. Less than the energy generation from a 14-carbon fatty acid
- D. In between the energy generation from a 14-carbon fatty acid and an 16-carbon fatty acid

24-Glycerol released from adipose tissue by hydrolysis of triglycerides is mainly.....?

- A. Excreted from the body
- B. Taken up by extra-hepatic tissues
- C. Re-utilized in adipose tissue
- D. Taken up by liver

25-Which of the following is correct about liver enzymes?

- A. ALT elevations persist longer than do AST.
- B. In chronic cases, ALT is more elevated than AST.
- C. ALT is more liver specific than AST.
- D. In the mild hepatocellular damage: AST/ALT > 1

26-The key regulatory enzyme of cholesterol synthesis is HMG Co-A reductase. All of the following inhibit the enzyme except

- A. Cortisol
- B. Insulin
- C. Statins
- D. Glucocorticoides

27-The process by which bile acid sequestrants lower serum cholesterol level is?

- A. Promote cholesterol excretion
- B. Divert cholesterol metabolism towards bile acid formation
- C. Inhibit cholesterol synthesis
- D. Promote cholesterol catabolism

28-Which of the following compounds is a member of the electron transport chain?

- A. NADH
- B. Cytochrome c
- C. Palmitoyl carnitine
- D. Carnitine

29-All of the following are correct about bile acids production, except?

- A. Is produced in liver
- B. Increased secretion of cholesterol into bile causes gallbladder stones
- C. Expression of 7- α -hydroxylase is upregulated by bile acids
- D. 7- α -hydroxylase is the rate-limiting step

30-Which of the following mechanisms would lead to the inhibition of cholesterol biosynthesis?

- A. Inhibiting the formation of HMG-CoA
- B. Preventing the reabsorption of bile acids from the intestine
- C. Increasing the conversion of cholesterol to bile acids
- D. Inhibiting the absorption of cholesterol from the diet

31-In persons with diabetes, excess acetyl-CoA produced by the liver, cannot be used by the TCA cycle, it accumulates in their blood circulation as?

- A. Glucose
- B. Acetoacetate
- C. Lactate
- D. Acetyl-CoA

32-What would be the consequence of carnitine consumption as a supplement?

- A. Accumulation of fat droplets in liver and muscle
- B. Increase in fatty acid synthesis in the liver
- C. Increased transportation of fatty acids into the mitochondria
- D. Increase in blood glucose concentration

33-Which of the following is correct about introducing double bonds into fatty acyl chains?

- A. Trans double bond is inserted
- B. Δ^9 double bond is inserted
- C. Elongase is responsible for the reaction
- D. RER is responsible for desaturating LCFA

34-Which of the following statements about the long-term fasting is correct?

- A. Adipose tissue can produce ketone bodies from its fat stores
- B. The liver use ketone bodies to produce energy
- C. Muscles produce ketone bodies from fatty acids
- D. Muscles and brain use ketone bodies to produce energy

35-A patient presents initial stage of myocardium infarction. Which of the following tests are more useful for diagnosis?

- A. CKMB- AST- LD4,5
- B. LDH1& 2 – AST - CKMB
- C. ALP-ALT-GGT
- D. GGT- AST- ALT

36-Steatorrhea is the presence of excess fat in feces. All of the following causes Steatorrhea except

- A. Gallbladder stones
- B. Short bowel syndrome
- C. Cystic fibrosis
- D. Diabetes

37-If all potential sources of ATP production are taken into account, the net number of ATP molecules formed per molecule of glucose in aerobic glycolysis is

A. 36

B. 2

C. 18

D. 6

38-Which one is an indicator for jaundice in plasma?

A. Ammonia

B. Creatinine

C. Urea

D. Bilirubin

39-A 7-year-old boy suffers from mental retardation and self-mutilation and has increased levels of serum uric acid. The symptoms are characteristic of Lesch Nyhan syndrome. Which of the following defective?

A. Salvage pathway for pyrimidine biosynthesis

B. De novo synthesis of pyrimidines

C. Xanthine oxidase

D. HGPRT

40-The ammonia from proteins degraded in muscle tissue is brought in the blood stream to the liver mainly via?

A. Glutamine

B. Alanine

C. Pyruvate

D. Glutamate